KWAME NKRUMAH UNIVERSITY OF SCIENCE AND TECHNOLOGY, KUMASI

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DEPARTMENT OF MATHEMATICS

Elliptic Curve Cryptography.

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A THESIS SUBMITTED TO THE DEPARTMENT OF MATHEMATICS, KWAME NKRUMAH UNIVERSITY OF SCIENCE AND TECHNOLOGY, KUMASI, IN PARTIAL FULFILLMENT OF THE REQUIREMENT FOR THE DEGREE OF BSc MATHEMATICS

Declaration

We hereby conscientiously declare that this dissertation is our own work under the able supervision of Mr. R.K Boadi, towards the award of the Bachelor of Science(BSc.) in Mathematics and that, to the best of our knowledge, it contains no material previously published by another person nor material which had been accepted for the award of any other degree of the university, except where due acknowledgement had been made in the text.

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Dedication

Acknowledgement

We are most grateful to the Almighty God for His mercies and guidance. For His Grace by which we can still hope.We are profoundly grateful to Dr. R.K Boadi (our project supervisor) we would like to show great appreciation for his devotion, guidance, support and encouragement. We are truly indebted to SPECIAL PERSON HERE for the enormous amount of time he spent to provide guidance, assistance and helpful constructive critique to our work.And to all the lecturers in the Department of Mathematics, KNUST, we say God bless you.

Abstract

Koblitz and Miller introduced the use of elliptic curves in encryption in 1985. The function of elliptic curves in cryptography was investigated.In this thesis we explore the basic implementation and security issues of several elliptic curve cryptosystems.Some of the aspect that will be addressed includes;Using a single solution to obtain many elliptic curve solutions,finding points algebraically,Application of rational numbers on elliptic curves in cryptography and how well programming codes are used to find private key cryptography.

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Chapter 1

INTRODUCTION

The theory of elliptic curves combines Algebra, Geometry, Analysis, and Number theory in a unique way. Elliptic Curve Cryptography emphasizes this interaction as it explores the underlying theory, allowing the reader to broaden their understanding of modern mathematical structures and their applications. Knowledge of topics from Abstract Algebra, such as Groups and Fields, as well as other key information on cryptography, are required to fully comprehend the two.

An elliptic curve is defined as a set of zeros of a cubic polynomial in two variables. If the polynomial includes rational coefficients, a description of the zeros or roots whose coordinates are integers or rational values can be requested. Rational points on Elliptic Curves is primarily concerned with this numerical theoretic question. Without knowing Finite Fields, it is often impossible to grab the meaning of the elliptic curve, cryptography, and its algorithms. What role do Finite Fields play in elliptic curves and cryptography? Finite fields can give the exact arithmetic to cryptosystems, which are required to carry out arithmetic operations without error.

1.1 Background of the study

Cryptography is the art and science of hiding messages in order to add secrecy to information security.(write the full definition here and if possible site someone here too who gave that dif.)

The term "cryptography" was created by combining two Greek words: "Krypto," which means "Hidden," and "Graphene," which means "Writing," along with the art of writing; cryptography is thought to have begun. Humans were organized into tribes, clans, and kingdoms as civilizations progressed. Power, fights, supremacy, and politics emerged as a result of this. These concepts stimulated people's natural need to interact privately with a select group of people, ensuring that cryptography continued to evolve.

Cryptography has its origins in Roman and Egyptian civilizations. The Oldest Cryptographic Technique is Hieroglyph(explain further the term). The usage of 'Hieroglyph' is the first known

indication of cryptography. The Egyptians employed hieroglyphic messages to communicate about 4000 years ago. Only the scribes who used to send communications on behalf of the kings were aware of this code.

Years after, Diffie and Hellman's(site their paper here) development of public key cryptography in 1976 not only transformed the science of cryptography, but it also had a significant impact on the path of computational number theory research. For the first time, the relative complexity of diverse number-theoretic tasks became a matter of practical importance.
The Rivest-Shamir-Adleman(RSA) cryptosystem, which was introduced in 1978, was the first viable public key system. It sparked a flood of research into the twin topics of factoring and primality testing. A second line of research in computational number theory arose from a type of public key cryptography based on the discrete counterpart of the logarithm function.
Discrete Log Cryptosystems(DLC) have not been in the public eye as long as RSA, but they have piqued the interest of researchers. The practical issues that have developed in discrete log cryptography have sparked a lot of research into the structure.

Neal Koblitz and Victor Miller(site their paper here) independently proposed the usefulness of elliptic curves in encryption in 1985. Since then, Elliptic Curve Cryptography (ECC) has grown into a substantial branch of Public Key Encryption.

Public Key Cryptography(PKC) systems are a type of encryption that employs the use of public keys. In a PKC system, different keys are used to encrypt and decode data. Since one of the keys in a PKC system is made public, and the system's security is determined by the key size.

Prior to PKC, mathematical problems were solved using prime factorization and discrete logarithm.

Elliptic curve cryptography has proved that it can provide the same level of security with smaller key sizes. Elliptic curve cryptography focuses on the security aspect of application systems for the most part. These systems have limitations in storage, computing performance, CPU and GPU architecture.

Cryptography has been useful from the past and it is still now. Cryptography has undergone series of evolution that has led to Cloud Computing, Data Security Issues, Challenges in the Cloud, DNA Computing and DNA Cryptography.

Cloud computing provide pervasive, easy, on-demand network connectivity, to a configurable pool of computational resources; of which include network infrastructures, servers, on & off storage sites, applications, and services. All these can be accessed and released quickly and with low effort and interaction with service providers or management effort.

1.2 Problem Statement

Elliptic curve cryptography has many benefit that has helped in the field of science and technology in terms of information secrecy and keeping of confidential data to government and institutions nevertheless it does have it negative aspect,Channel attacks and twist-security attacks are examples of such attacks. Both types try to compromise the ECC's private key security.This project seeks to address and highlight some possible solutions to these problem.

1.3 Objectives of study

This project's major goal is to better understand elliptic curves and how they relate to cryptography. Abstract algebra concepts such as groups and fields are examined first as a prerequisite for the thesis.

Other Objectives are;

- To understand how private key is generated and the attacks on security.
- To examine cryptographic application of elliptic curves.

1.4 Significance of the study

Elliptic curves can be found in a wide range of mathematical fields,that is from number theory to complex analysis. Elliptic curves can be used for encryption, digital signatures, pseudo-random generators, and a variety of other applications. They are also employed in a number of integer factorization methods with cryptographic applications, such as the Lenstra elliptic-curve factorization, illustrating the importance of elliptic curves in human lives in the 1980s and years after.This thesis would illustrate how to employ elliptic curves in various applications particularly when they are defined over finite fields. It also examines how and why such curves are useful in the field of cryptography.

1.5 Scope of the study

This thesis is not intended to address a specific topic or to produce a novel finding; rather, as previously said, it intends to provide a grasp of elliptic curves, how it came to being, and how they are applied in elliptic curve cryptography. The subject of elliptic curves is a strong blend of abstract algebra in particular group law and fields, as we will see in this paper.

At the moment, elliptic curves can be used for encryption, digital signatures, pseudo-random generators, and a variety of other applications. They are also employed in a number of integer factorization methods with cryptographic applications.Elliptic curves are employed in cryptographic protocols like Bitcoin and Austrian smart e-ID, to name a few examples. We will go over each of these applications.Elliptic curves are a common theme in algebraic geometry and cryptography, among other branches of mathematics.We can find number theory problems in the history of elliptic curves. Theoretical complex functions.The elliptic curve theory is, in fact, a combination of the two.

.....As a survey of the history of elliptic curves, we look at the 3rd degree Diophantine equations and the congruent number problem as examples of number theoretic trails of elliptic curves. We also look at elliptic integrals and elliptic functions, which give us an idea of where the name "Elliptic Curve" came from. We show how the solution of 3rd degree Diophantine equations and elliptic functions are connected to elliptic curve cryptography.....

1.6 Organization of Study

This thesis is divided into five chapters, the first of which provides a broad summary of the entire thesis as well as the circumstances of the research. This chapter begins with an introduction, the study's history, a problem statement, and the study's goal.We outline the works connected to Elliptic curve cryptography in the project's second chapter.The third chapter contains thorough information in elliptic curve cryptography.In chapter four,you will find detailed solutions and information and also the generation of the private key cryptography.In chapter five,the solutions and discussions has led to a conclusion and recommendations.

Chapter 2

LITERATURE REVIEW

Narayan (2018a) presented the order of the related algebraic group was significantly easier to determine than in the general case for a family of elliptic curves for cryptographic purposes.This made it easier to meet the cryptographic requirement of a big prime factor for this arrangement. Another merit of this family was that group operations are slightly simplified.

Narayan (2018b) presented a detailed overview of cryptography method that was based on complex mathematical problems such as prime number factorization and the discrete logarithm problem for elliptic curves. The method of continuously adding a point to an elliptic curve was known as elliptic curve point multiplication. It is used to create a trapdoor function in Elliptic Curve Cryptography (ECC).

Koblitz et al. (2000) looked over the growth of elliptic curve cryptosystems.In 1976 Diffie and Hellman introduced the public-key cryptography.It had been realized that the discrete logarithm problem had the potential to be used in public-key cryptosystems. Diffie and Hellman described the discrete logarithm problem as the challenge of finding logarithms with regard to a generator in the multiplicative group of the integers modulo a prime, the discrete logarithm problem had since been redefined.This idea could be widened to arbitrary groups and, in particular, to elliptic curve groups.The resulting public-key systems had a small block size, are fast, and secure.

Koblitz (1987) discussed Public key cryptosystems that used the multiplicative group of a finite field had elliptic curves over finite fields.This is because the elliptic curve equivalent of the discrete logarithm problem was anticipated to be harder than the conventional discrete logarithm problem, especially over $GF(2^n)$, these elliptic curve cryptosystems may be more safe on the nonsmoothness of the order of the cyclic subgroup created by a global point.

Smart (1999) proposed a basic technique that addressed the discrete logarithm issue on trace one elliptic curves,which led to a linear algorithm. In practice, this meant that when elliptic curves were selected for cryptography, all curves with group orders equal to the finite field's order must be eliminated.

Barbulescu et al. (2015) investigated a field $GF(p_n)$, where n was a significant integer bigger that 1, the hardness of the discrete logarithm issue was examined. The difficulty of this challenge was at the center of security evaluations for torus-based and pairing-based encryption, while being less studied than the tiny characteristic case or the prime field situation. The Number Field Sieve(NFS) was the most well-known approach for tackling this problem. The ability to identify appropriate polynomials that specified the extension fields used in NFS was a significant component of this technique. We provided two new techniques for this problem, each of which modifies the asymptotic complexity and allows for record-breaking calculations. It demonstrate these findings by computing discrete logarithms over a field $GF(p_2)$ with cardinality of $180digits(595bits)$. Chen (2008) looked at how the Legendre symbol behaved when applied to rational points on an elliptic curve, and came up with a family of binary sequences that had string pseudorandom features. That was, both the well-distribution measure and the correlation measure of order k of the generated binary sequences are "small" as measured by estimate of particular character sums along elliptic curves.It also offered a lower constraint on the linear complexity profile of these sequences.Our findings suggest that such sequences behave similarly to Legendre sequences.

Koblitz et al. (2000) reviewed the growth on elliptic curve cryptosystems,Elliptic Curve Cryptography has seen increasing success in security applications in recent years due to features such as a small key size with a high level of protection. Their popularity has resulted in a wide range of implementations in terms of algorithms, curves, coordinate systems, platforms, and so forth.

Vidhya et al. (2019) highlighted works that overcame the drawback of the GCD attack in RSA. One of the most important varieties of Asymmetric cryptography was Rivest, Shamir, and Adleman (RSA). When compared to other cryptography techniques, the ky size in RSA is larger. Any two prime numbers were used to create the keys, with the Greatest Common Divisor (GCD) attacks, an unauthorized user might simply obtain the keys . One of RSA cryptography's major flaws was with a small key size,Elliptic Curve Cryptography (ECC) is also asymmetric cryptography. The goal of the proposed effort is to solve the GCD attack RSA flaw. Employing an Exclusive OR(XOR), the RSA private and public keys were coupled with the ECC private and public keys, and new hybrid private and public keys were created. A new hybrid private key could be used to decrypt RSA data. The suggested method has the advantage of creating three tiers of cyptographic keys, which can avoid the GCD attack in RSA. When compared to the RSA algorithm's straight forward key creation, the key strength

is greater.

Sutikno et al. (1998) discussed the implementation of elliptic curves cryptosystem, After Diffie and Hellman introduced public key cryptography, T. ElGamal introduced a discrete logarithm problem-based public key cryptosystem approach in 1985. Neil Koblitz and Victor Miller proposed elliptic curve cryptosystems for the first time in 1985, on their own.This is because they used elliptic curve groups for arithmetic, elliptic curve cryptosystems are unique.The discrete logarithm issue in a set of points on an elliptic curve formed over a finite field is the basis for this cryptosystem. In an elliptic curve group, the discrete logarithm issue looked to be significantly more difficult than in other groups.

Schroeppel et al. (1995) explained how to build the Diffie-Hellman key exchange algorithm using a set of points on an elliptic curve over the field F_{2n}. A software version of this that used $n = 155$ might be optimized to achieve slightly quicker computation rates with a similar level of security than non-elliptic curve versions. The highly eddicient implementation was made possible by F_{2n} quick calculation of reciprocals.

Kumar et al. (2016) proposed a novel image security algorithm based on Elliptic Curve Cryptography (ECC) and DNA encoding, with the incressed usage of media in communications, image encryption was required to protect against attacks. The RGB image was first encoded using DNA encoding, then asymmetric encryption using Elliptic Curve Diffie-Hellman Encryption was applied (ECDHE). For analysis, the suggested technique was used to standard test photos. Key spacing, key sensitivity, and statistical analysis were all examined. According to the findings, the suggested method can withstand exhaustive attacks and is suitable for real applications.

Bellare et al. (2000) addressed the security of public-key cryptosystems in a “multi-user” scenario, specifically in the presence of assaults utilizing the encryption of related messages using distinct public keys, as demonstrated by Hastad‘s classical RSA attacks. We demonstrated that security in a single -user context entails security in a multi-user setting if the former is regarded in the broad sense of “indistinguishability”, pinning down a number of methods that are guaranteed to be secure against hastad-type assaults.

ElGamal (1985) offered a novel signature technique, as well as a Diffie-Hellman key distribution scheme implementation that resulted in a public key cryptosystem. The difficulty of

computing discrete logarithms over finite fields provided the security for both systems.

Mahto et al. (2016) described data transmission technologies that were transformed, as a result of this.It had provided platform for communicative parties to swiftly communicate information while simultaneously providing an opportunity for adversaries to strike.Cryptographic techniques were used to guarantee secrecy, integrity, and authentication to insecure data at rest,compares and contrasts the security of two widely used public-key cryptography systems,Rivest Shamir Adleman (RSA) and Elliptic Curve Cryptography (ECC).Rivest Shamir Adleman (RSA) was the first-generation public-key cryptography that was immensely popular since its introduction, while ECC has only lately gained prominence. The Integer Factorization Issue (IFP) was used to secure the RSA cryptosystem, while the elliptic curve discrete logarithm problem (ECDLP) was used to secure the ECC cryptosystem.The major merit of ECC over RSA was because the best-known technique for solving the elliptic curve discrete logarithm problem (ECDLP) requires full exponential time, whereas RSA's (IFP) takes sub-exponential time. This means that ECC can employ far fewer parameters than RSA while maintaining the same level of security.For instance the RSA technique requires a key to produce 112 bits of protection.ECC requires a key size of 224-255 bits, while RSA requires 2048 bits.

Danger et al. (2013) summarized elliptic curve encryption is subject to side-channel attacks. These attacks took use of biases in a variety of leakages, including power consumption, electromagnetic emission, execution time, and so on. To counter known assaults, countermeasures must be integrated. Because no one countermeasure can protect against all types of attacks, many of them must be integrated. However, because each has a hefty price tag, they can not all be used at the same time. Depending on the situation and the trade-off between security and performance, it is crucial to choose countermeasures carefully. The side-channel attacks on Elliptic Curve Cryptography are summarized in this work, along with countermeasures.The cost in terms of time and space is stated for each countermeasure. Some assaults, such as the doubling attack, are clarified; others, such as the horizontal SVA, are enhanced; and new attacks, such as the horizontal attack against unified formulae, are detailed.

Chapter 3

Methodology

Elliptic curve cryptography was discovered in 1985 by Victor Miller and Neil Koblitz as an Elliptic Curve Method (ECM) was applied on cryptography known as Elliptic Curve Cryptography. Cryptography based on Elliptic Curves (ECC) has the unique feature of being the finest up to now.It can be solved using a well-known approach that takes full exponential time to complete.The Elliptic Curve Logarithm is responsible for its safety.In a group specified by the Discrete Logarithm Problem (DLP) over a finite field, using points on an elliptic curve,To attain the same level of security as traditional public key cryptography techniques, the required key size must be drastically reduced.As a result of that,an appropriate public key scheme would be one that is cost-effective in terms of compute and key sizes,Compared to other public key cryptosystems, ECC offers the highest strength-per-bit. The smaller the key, the less bandwidth, memory, and processing power are used.

3.1 Group Law

We first consider the case of Weierstrass elliptic curves. Let L be some line. We make the following rules.

1. The identity element is $0 = [0 : 1 : 0]$.
2. If A, B, C are 3 distinct points of $L \cap E$ then A + B + C = 0.
3. If $L \cap E$ consists of exactly 2 points A and B, and L is tangent to E at A, then A + A + B = 0.
4. If $L \cap E$ is just a single point A and L is tangent to E at A, then we have A + A + A = 0.

   It is commonly known that when the points on an elliptic curve are added together, they form an abelian group. Let E be the Weierstrass equation for an elliptic curve.

3.1.1 Weierstrass Equation

Weierstrass equation is given as;

$$Y^2 = X^3 + aX + b \tag{3.1}$$

The addition rule is given below: For all P, Q, E

(i) O+P = P and P+O = P. (So O serves as the identity element.)

(ii) -O = O.

(iii) If P = (x_1, y_1) :f: 0, then -P = $(x_1, -y_1 - a_1x_1 - a_3)$. (Note that P and - P are the only points on E with x-coordinate equal to X_1

(iv) If Q = -P, then P +Q =O.

(v) If $P \neq O, Q \neq O, Q \neq -P$, then let R be the third point of intersection (counting multiplicities) of either the line $\overline{PQ}$ if $P \neq Q$, or the tangent line to the curve at P if P = Q, with the curve (as usual, the tangent line to the curve f(x, y) = 0 at P = (a,b) is the line $\frac{\partial f}{\partial x}(P)(x-a) + \frac{\partial f}{\partial y}(P)(y-b)$= 0. Then P+Q = -R

3.1.2 Group Properties

Elliptic curves are the set of points that are obtained as a result of solving an elliptic curves function over a predefined space.

Definition::

An elliptic curve over K is a nonsingular projective algebraic curve E of genus 1 over K with a chosen base point o in E

Some points in elliptic curve:

• Elliptic curves have point with coordinated in any finite field that is rational point, real point, complex point and finite point.

• Elliptic curves with point with point in Fp are finite group.

3.2 Using one solution to find various solution to an elliptic curve

Elliptic curve defined by a certain type of cubic equation with two variables. The set of rational solutions to the equation has an extremely interesting structure including the group law

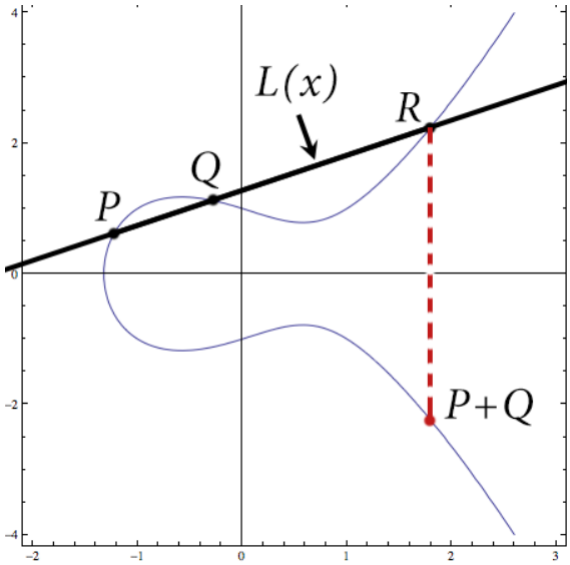

Figure 3.1: Geometric figure of an Elliptic curve

Taking the figure below:

We are to find all the point of on the figure below,and doing so we first find the values of a

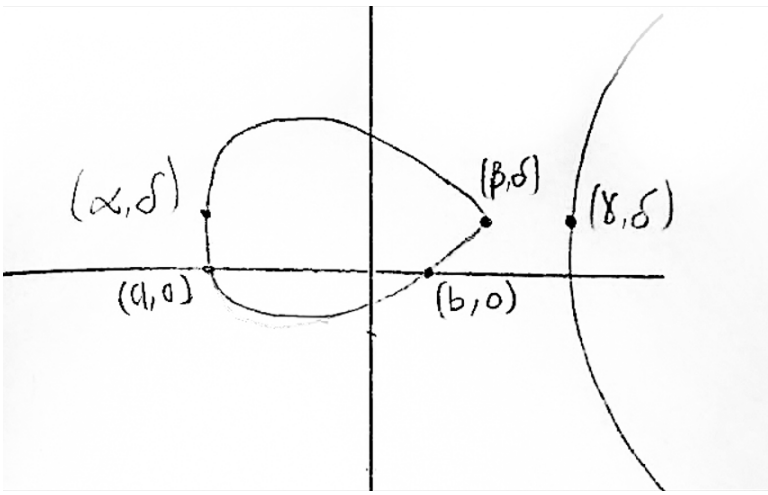

Figure 3.2:

and b

Finding the value of a and b

Given:

Let $Y = 0$

$0^2 - 0 = x^3 - x$

$x(x+1)(x-1) = 0$

From here, $a = -1$ and $b = 1$

Find the values of δ

Adding $\frac{1}{4}$

$Y^2 - Y + \frac{1}{4} = X^3 - X + \frac{1}{4}$

$(Y - \frac{1}{2})^2 = X^3 - X + \frac{1}{4}$

From $(Y - \frac{1}{2})^2 = X^3 - X + \frac{1}{4}$,

$\delta = \frac{1}{2}$

Why the curve is symmetric about Y=δ

$(Y - \frac{1}{2})^2 = X^3 - X + \frac{1}{4}$

$Y = \frac{1}{2} \pm \sqrt{X^3 - X + \frac{1}{4}}$

Note: if $\left(X, \frac{1}{2} + \sqrt{X^3 - X + \frac{1}{4}}\right)$ is on the curve, so is the point $\left(X, \frac{1}{2} - \sqrt{X^3 - X + \frac{1}{4}}\right)$.

Finding a cubic equation with roots α, β, γ

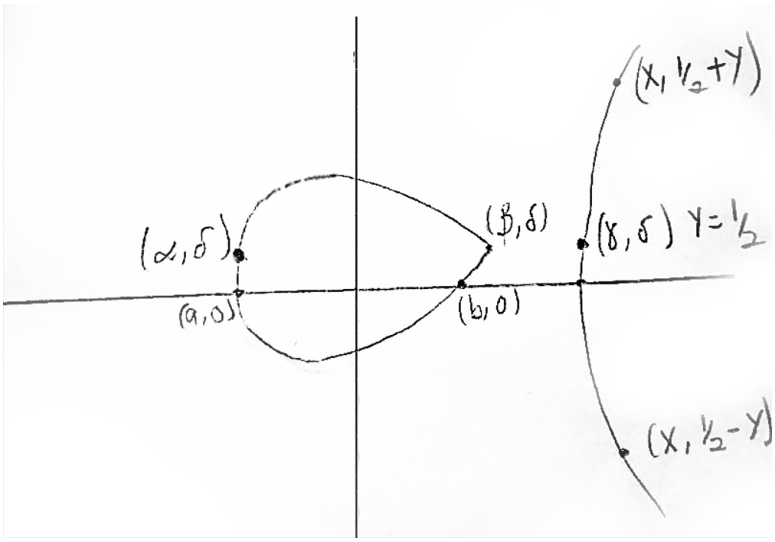

Figure 3.3:

Taking the second line obtained from the object to be our x-axis

Let $Y = \frac{1}{2}$

$\left(\frac{1}{2} - \frac{1}{2}\right)^2 = X^3 - X + \frac{1}{4}$

$X^3 - X + \frac{1}{4} = 0$

Find $\alpha + \beta + \gamma$

$X^3 - X + \frac{1}{4} = (X - \alpha)(X - \beta)(X - \gamma)$

$X^3 - X + \frac{1}{4} = X(X^2 - (\alpha + \beta)X + \alpha\beta(X - \gamma)$

$X^3 - X + \frac{1}{4} = X^3 - (\alpha + \beta + \gamma)X^2 + ...$

$X^3 - X + \frac{1}{4} = -(\alpha + \beta + \gamma) = 0$

$\alpha + \beta + \gamma = 0$

Let C be the circle with (α, δ),(β,δ) as ends of a diameter.

Equation of a circle: $a^2 + b^2 = r^2$

Put the centre $\left(\frac{\alpha+\beta}{2}, \frac{1}{2}\right)$ into the equation of a circle

Centre $\left(\frac{\alpha+\beta}{2}, \frac{1}{2}\right)$

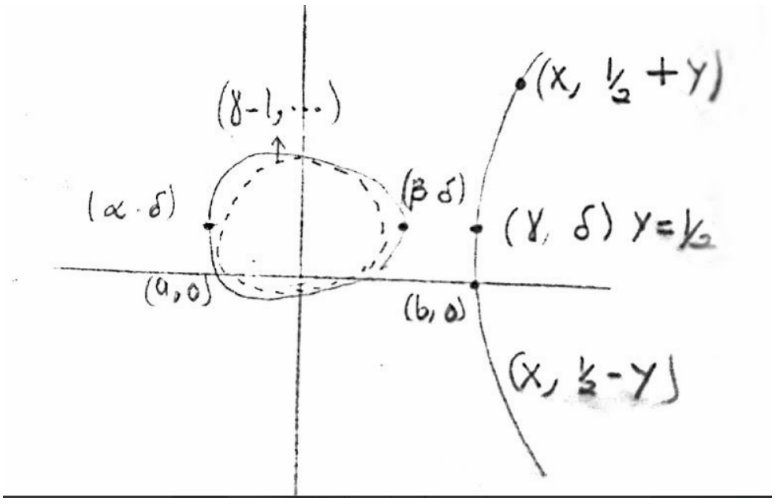

Figure 3.4:

$\left(X - \frac{\alpha+\beta}{2}\right)^2 + \left(Y - \frac{1}{2}\right)^2 = \left(\frac{\beta-\alpha}{2}\right)^2$

Showing that C intersect S at two other points and finding the X-coordinate in terms of δ

$\left(X - \frac{\alpha+\beta}{2}\right)^2 + X^3 + X - \frac{1}{4} = \left(\frac{\beta-\alpha}{2}\right)^2$

Note we know that α and β are solutions to $\left(X - \frac{\alpha+\beta}{2}\right)^2 + X^3 + X - \frac{1}{4} = \left(\frac{\beta-\alpha}{2}\right)^2$,let S be the 3rd solution

Since the coefficient of X^2 in $\left(X - \frac{\alpha+\beta}{2}\right)^2 + X^3 + X - \frac{1}{4} = \left(\frac{\beta-\alpha}{2}\right)^2$ is 1

So $\alpha + \beta + S = -1$

$-\gamma + S = -1$

S=γ-1

3.3 Finding point algebraically

For any points P and Q on an elliptic curve, EC group law point addition can be used geometrically to find various point on the plane.If we draw a line through P and Q,this line will intersect the elliptic curve at a third point(-R). The reflection of this point about x-axis, R is the addition of P and Q.

Bezout's Theorem establishes that the intersection of P and Q is valid and it state's; Suppose that X and Y are two plane projective curves defined over a field F that do not have a common component (X and Y are defined by polynomials whose greatest common divisor is a constant). Then the total number of intersection points of X and Y with coordinates in an algebraically closed field E which contains F, counted with multiplicity, is equal to the product of the degrees

of X and Y . This makes the formation of the third point R after the intersection of P and Q.

Properties

1. If a line intersects two points,it intersects a third
2. If a line is tangent to the curve, it intersects another point
3. All vertical lines intersect the point at infinity
4. The curve has an ever-increasing slope after a point of inflection

The Addition Law

The curve E under point of addition is an abelian group with these properties:

1. Identity-at infinity
2. Inverse-the reflection of a point.
3. Associativity
4. commutativity

First step in finding the equation is to define the line.

$P = (X_1, Y_1)$ and $Q = (X_2, Y_2)$

• The point-slope from the line.

$\lambda = \frac{y_2 - y_1}{x_2 - x_1}$

$y - y_1 = \lambda\ (x - x_1)$

• The slope-intercept form of the line.

$y = \lambda$x - $\lambda\ x_1 + y_1$

$\beta = y_1$ - $\lambda\ x_1$

$y = \lambda$x+β

Finding x_3 and y_3 of points 3

y=$\lambda\ x + \beta$

squaring both sides

$y^2 = (\lambda$x+$\beta)^2$

The curve $y^2 = x^3 + ax + b$

$(\lambda$x+$\beta)^2 = x^3 + ax + b$

$0 = x^3 - \lambda^2 x^2 - 2\lambda x\beta - \beta^2$+ax+b

x_1,x_2,x_3 *are the roots.*

$(x_1 + x_2 + x_3) = \lambda^2$

$x_3 = \lambda^2 - x_1 - x_2$

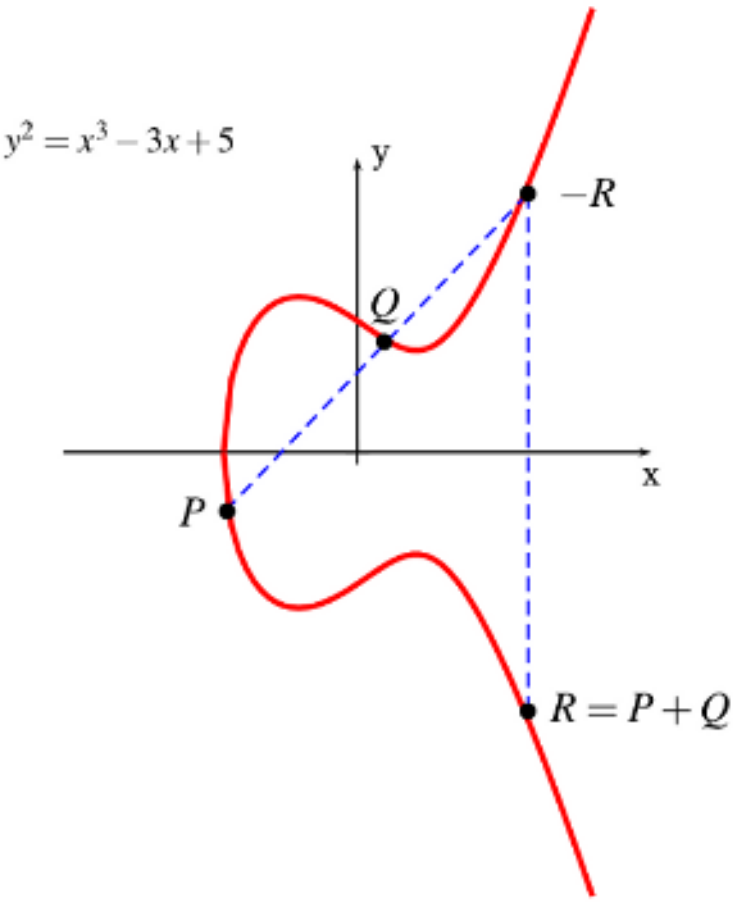

Figure 3.5:

$y_3 = \lambda(x_1 - x_3) - y_1$

3.3.1 Identity

Definition:

Let P be the elliptic curve E, and O be the point at infinity,then P+O=O + P Finding the point at infinity 0

1. Adding infinity results in a vertical line.
2. Then find the third point of intersection.
3. After the reflection the result turns to be the original point.

$P + \infty = P$

The figure below illustrate this concept:

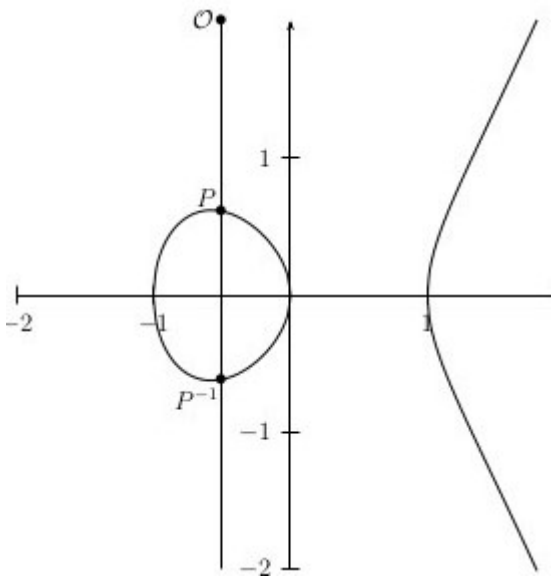

Figure 3.6:

3.3.2 Inverse

An operation (+) is said to be invertible within a set if each element P has an element with the same set P^{-1} such that $P + P^1 = O$ where O is the identity of the operation within the set.In this case,P and P^{-1} are said to be inverses of each other

P+$-P$=∞

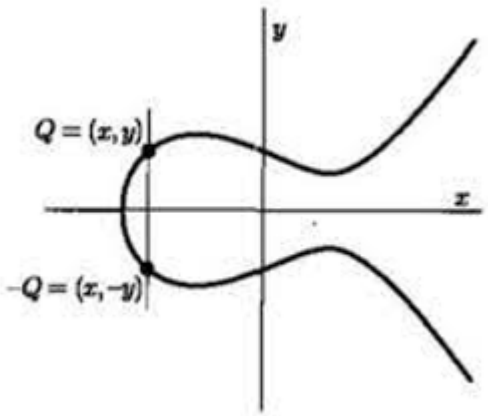

Figure 3.7:

The reflection of a point is it inverse

so, if we define

P=(x,y) and $-P$=(x,$-y$)

$P - P$=∞

3.3.3 Associative

An operation is said to be associative if $P + (Q + R) = R + (P + Q)$ for all P,Q,R within a set.The steps used to find the operation of normal addition (+) on associativity on a geometry is below:

1. Taking a line through P and Q we get $P + Q$ and it reflection is $P + Q$.
2. Taking a line through P and Q we get $P + Q$ and it reflection is $P + Q$.
3. Taking a line through Q and R we get $Q + R$ and it reflection is $Q + R$.
4. Taking a line through $Q + R$ and P it turns $P + (Q + R)$.
5. Taking a line through $P + Q$ and R it turns $R + (P + Q)$.
6. The line $R + (P + Q)$ and $P + (Q + R)$ intersect at a point whose reflection form a vertical line thus at ∞.
7. This makes $R + (P + Q)$ and $P + (Q + R)$ equal. Taking a line through Q and R we get $Q + R$ and it reflection is $Q + R$.
8. Taking a line through $Q + R$ and P it turns $P + (Q + R)$.

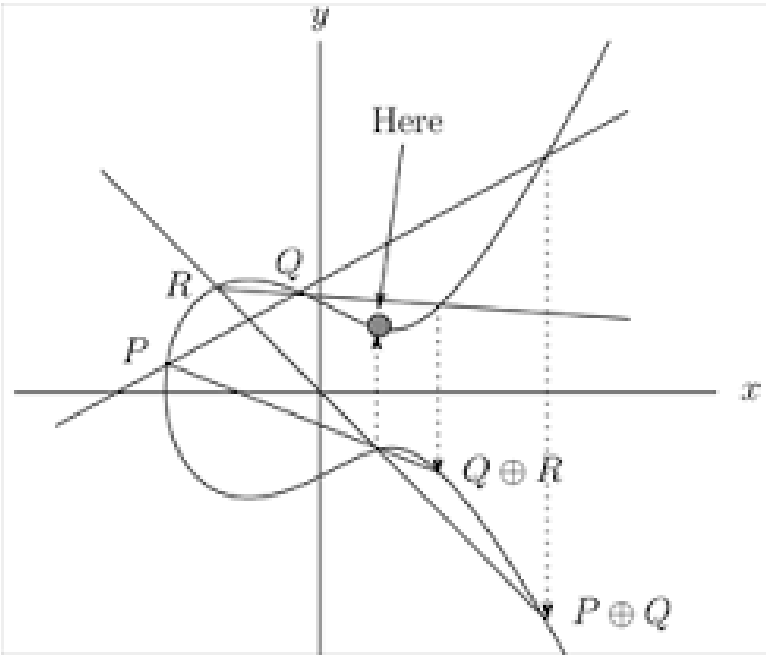

Figure 3.8:

9. Taking a line through $P + Q$ and R it turns $R + (P + Q)$.

10. The line $R + (P + Q)$ and $P + (Q + R)$ intersect at a point whose reflection form a vertical line thus at ∞.

11. This makes $R + (P + Q)$ and $P + (Q + R)$ equal.

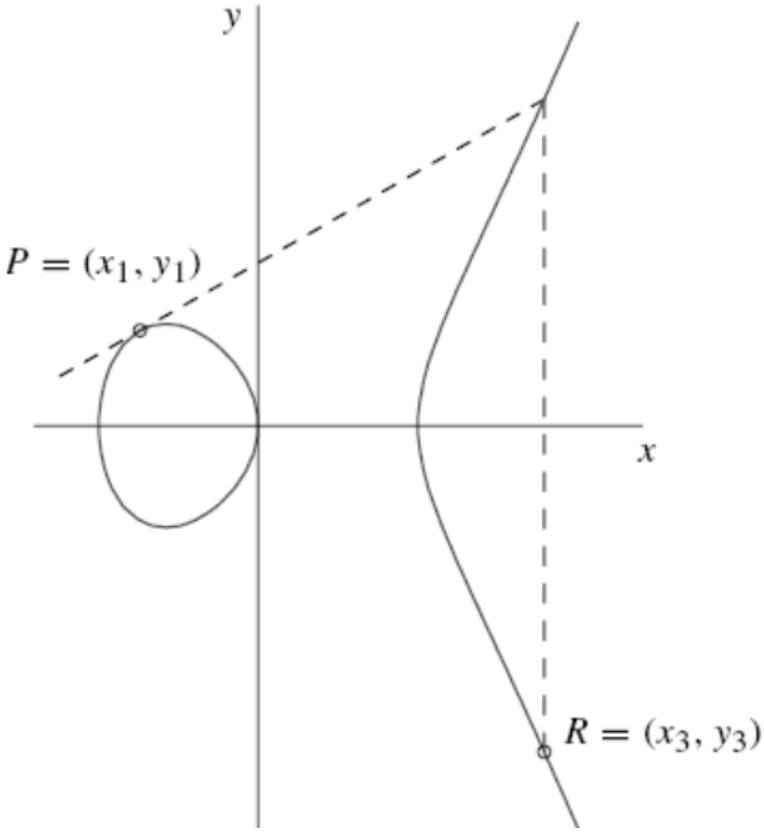

Figure 3.9:

3.3.4 Point Doubling

When the points P and Q are with the same coordinates, addition is identical, except there is no well-defined straight line between P and Q, so the operation is closed using the limiting case, the tangent to the curve, E, at P and Q.

Point doubling is calculated using following equation:

$$2P = (x_3, y_3)$$

,where

$$x_3 = \lambda^2 - 2x_1$$

$$y_3 = \lambda(x_1 - x_3) - y_1$$

$$\lambda = \frac{3x_1^2 + a}{2y_1}$$

Thus:if $P = Q$

The properties of the group law all works together for good for elliptic curve cryptography.It basically works good under the addition law and point doubling,in elliptic curve cryptography.This makes the security of elliptic curve cryptography more reliable because series of generator point has to be traced back to the original point and it will be very difficult to do so based on the repeated point addition and point doubling.

3.4 cyclic group

Let G be a finite group of order n is called cyclic if there is an identity element $a \in G$ such that the cyclic subgroup generated by a is the entire group G. In other words, $G = a^n : n \in Z$. Such an element a is called a generator of G.
Note: A cyclic group is abelian but a group that is abelian is not necessarily.
To do some work with the points of elliptic curves, you will need to know what an elliptic curve is and what a group law is. The cyclic subgroup, the base point, and the order can be determined using the points of an elliptic curve when they are restricted to finite fields of integers modulo a prime.

3.4.1 The Key Pair Generation of ECC

Elliptic curve cryptography requires the following parameters below:

1. The finite field's size is specified by the prime p.
2. The elliptic curve equation has coefficients a and b.
3. The cyclic subgroup is generated by the generator G.

4. n is of the subgroup's order.

The private key is a random integer d selected from the range $1, ..., n-1$, where n is the subgroup order. The public key is $PK = dG$, where G is the subgroup's generator. Finding PK is simple if we have d and G, as well as the other domain parameters. Finding the private key d is difficult if we know PK and G since it requires solving the discrete logarithm issue.

3.5 Diffie - Hellman Key Exchange on elliptic curve cryptography

Suppose Budu-Biney and his Bank establish a common key with his bank with the aim to exchange data as part of symmetric encryption scheme.

1. Budu-Biney and his Bank agree to use a specific curve (elliptic curve E) over a finite field Fq that makes the discrete logarithm problem impossible in E (F_q). They all agreed on a point $P \in (F_q)$ indicating that the P-generated subgroup has a big order.
2. Alice selects a secret $b \in Z$ and enumerate bP = Pb and sends Pb to his bank.
3. Bank select a secret c∈Z and enumerate cP = Pc and sends Pc to Budu-Biney
4. Budu-Biney enumerate $bPc = bcP$.
5. The Bank enumerate $cPb = bcP$. To extract a key from bcP, Budu-Biney and his bank utilize a widely accepted approach, such as the final 256 bits of bcP's x-coordinate.

On the elliptic curve E, the finite field F_q and the points P, bP, and cP.The only way an intruder can get Budu-Biney's secret key is to use P, bP, and cP to determine bcP. If the intruder can find the discrete logs over E(F), them the intruder could use P and bP to find Budu-Biney's secret key b.Then the third person could now find b(cP) to get bcP.Solving discrete logs knowledge is required to solve it and without it is assumed to be infeasible to fine over E(F_q).

To determine whether $Q = bcP$ given P, bP,$cP \in E(F_q)$ and a point $Q \in E(F_q)$ in a comparable problem known as the Decision Diffie-Hellman problem. In other words, can you verify that bcP is right if you are given it as a tip? In some situations, as shown below, the Weil pairing can be utilized to solve the Decision Diffie-Hellman issue.

3.6 ElGamal Public Key Encryption

Assume Budu-Biney wants to communicate with his Bank. To begin, the bank creates it public key in the manner described below. The bank selects an elliptic curve E over Fq that makes the discrete logarithm issue for E(Fq) impossible, as well as point $P \in E(Fq)$ P have a large prime order. the bank also selects a secret $a \in Z$ and calculates $B = aP$. As a result, Budu-Biney's public and private keys are E, Fq, P, and B, respectively.
To send a message to the bank, Budu-Biney follows the steps below.

1. The Bank's public key is downloaded.
2. Use a point $M \in E$ to convey her meaning (Fq).
3. He selects a secret $k \in Z$ and calculates $M_1 = kP$.
4. $M_2 = M + kB$ is calculated.
5. $M_1 and M_2$ are sent to his bank.

The bank calculates the cipher text in order to decrypt it. $M_2 - sM_1 = (M + kB) - s(kP) = (M + skP) - skP = M$
if the intruder has the bank's public key as well as the points $M_1 and M_2$, but there appears to be no way for the intruder to access M if the intruder does not know how to solve discrete logs. Budu-Biney must use a different $k \in Z$ each time he sends a message to the bank, which is an essential note about the $k \in Z$ he chooses at random.The intruder can quickly detect Budu-Biney's usage of the same integer k if he computes M and M0 with the same value of k. The bank may then use $M_2 M_{20} = M_0$ to recover the message M by remembering that $M_0 = MM_2 + M_2$

3.7 ElGamal Digital Signatures

Let's say Budu-Biney wants to sign a bank document electronically. One method is to digitalize Budu-Biney's signature; however, The bank can copy Budu-Biney's digital signature and then attach it to any document.The steps require to be used so that such a digital signature can only be used one time.However, someone should be able to verify that the signature is genuine and that Budu-Biney is the one who signed the document. One solution to the problem is based on the discrete logarithm problem's difficulty.Budu-Biney first picks an elliptic curve E over a finite field F_q in which the discrete logarithm issue is impossible to solve (Fq).He also chooses $A \in E(F_q)$ as a location where $|A| = N$ is a huge prime.Budu-Biney also selects $a \in Z$ secret

and calculates the location $B = aA$. Finally,He selects a function $f : E(F_q) \longrightarrow Z$ that has the property of having a huge picture and just a small number of inputs that result in a certain output.As a result, Budu-Biney's public key is composed of the letters $E, F_q, f, A, andB$. The secret $a \in Z$ is his private key. The order of point A, on the other hand, does not have to be made public. As a result, Budu-Biney does the following to sign the bank document:

1. The bank document is represented as $m \in Z$. If $m > N$, Budu-Biney will select a larger curve.

2. R = kA is solved by picking a random $k \in Z$ and setting $gcd(k, N) = 1$.

3. Solves $s \equiv k^{-1}(m - af(R))(modN)$

After then, the message is signed (m, R, s). It is worth noting that Budu-Biney is not interested in keeping the message m a secret. If he wants to do so, he will need to employ some type of encryption.The bank then confirms that the communication is from Budu-Biney by saying:

1. The bank downloads Budu-Biney's information

2. Calculates $V_1 = f(R)B + sRandV_2 = mA$.

3. If $V_1 = V_2$, then the bank declares that the signature is authentic.

R is the same in both signatures, thus you will have to do it twice. The following are the equations for s and $s^{'}$:

$$ks \equiv (m - af(R))(modN)$$

$$ks^{'} \equiv (m^{'} - af(R))(modN)$$

Taking the difference between these two match gives $k(s - s^{'}) \equiv (m - m^{'})(modN)$.Set $d = gcd(s - s^{'}, N)$ then there are d possible values of k.The intruder tries all of them until he finds R = kA. He can then locate an as previously once he knows k.

Chapter 4

According to Silverman and Tate (2015) To find an integer $c \in Z$ and search for Diophantine equation solutions.y^2 - x^3= c.Suppose we are looking for a solution in the rational numbers $x, y \in Q$.If (x, y) is a solution with rational x and y and $y \neq 0$, it is simple to verify that the pair is correct.

$$(\frac{x^{4-8cx}}{4y^2}, \frac{-x^6 - 20cx^3 + 8c^2}{8y^3})$$

is an answer to the identical equation in rational numbers.If the original solution is $x, y \neq 0$, then repeating the process will provide an unlimited number of different solutions.if an integer can be written using nonzero rational numbers as the difference between a square and a cube

4.1 Application of rational numbers on elliptic curves in cryptography.

A non-singular projective curve of genus one with a determined base point is called an elliptic curve. Put a field k, such as the rational numbers Q, a finite field F_q, or even a function field C(t), in fancier terminology for the inspired. An elliptic curve E defined over k is defined as the functor that connects fields K containing k to an algebraic set. This is Because we can make a linear change of variables, an elliptic curve is represented by a cubic equation of the form $y^2 = x^3 + Ax + B$ for some $A, B \in k$ fulfilling $4A^3 + 27B^2 \neq 0$ if k is not of characteristic 2 or 3. Indeed, on the curve with z = 0, $O = (0 : 1 : 0)$ is the only projective point $P = (x : y : z)$.

4.1.1 Fermat curves

First of all, suppose the curve $a^3 + b^3 = c^3$ over a field $k = Q$, By substitution $a = 32 - 8y$, $b = 40+8y$, and $c = 24x$, determine the elliptic curve $y^2+y = X^3-7$. The curve $3a^3+4b^3 = 5c^3$ on the other side, is not an elliptic curve over Q because it has no Q rational solutions other than the degenerate one, namely $a = b = c = 0$. However, this equation does define a genus one projective curve. For various exponents n, the curve $a^n + b^n = c^n$ over Q has piqued people's interest. When n is a large enough integer, Pierre de Fermat question if there were any

nonzero Q rational answers. When $n = 2$, any Pythagorean Triple, such as $(a, b, c) = (3, 4, 5)$, will suffice as a desirable answer. We have seen that when $n = 3$, elliptic curve arise, but this isn't the only exponent where this occurs. Surprisingly, every solution (a, b, c) provides a point $P = (x, y, 1)$ on the elliptic curve E when $n = 7$. So $E : y^2 + xy = x^3 - x^2 - 107x + 552$ by a hard and nontrival-substitution

$$x = 49\frac{(a^2 + b^2 + c^2 + ab + ac + bc)^2 + abc(a + b + c)}{(a + b + c)^4} - 12$$

,

$$y = \frac{7(x + 12)(a^2 + b^2 + c^2) - x(a + b + c)^2}{(a + b + c)^2}$$

By focusing on Q-rational P on E, one can one can identify all Q-rational solutions to $a^7 + b^7 = c^7$.

4.1.2 The Mordell Weil group

The structure of the abelian group $E(K)$ is a suitable topic for discussion. Louis Mordell's famous theorem for $K = Q$,for finite extensions $KofQ$, says that $E(K)$ is finitely produced, that is, $E(K) = E(K)torsZr$ for some finite group $E(K)tors$ consisting of torsion elements and some non-negative integer r termed the rank.

$$y^2 = x(x + d_1)(x + d_2)$$

Mazur's Theorem asserts that E(Q)tors (Z/2 Z) (Z/2 m Z) for some m = 1, 2, 3, or 4 for some m = 1, 2, 3, or 4. We can choose $T_1 = (0 : 0 : 1)$ as a generator of order 2, leaving only T_2 as a generator of order 2 m to discover.The rank is much more difficult to determine: there is an injective group homomorphism, which is based on a Mordell notion.

4.2 The Discrete Logarithm Problems

There are a few unsolved issues in the field of discrete logarithms in cyclic groups. The issues arose after the Diffie-Hellman protocol was introduced in 1976. These unsolved issues are listed below.

1. The Problem of Discrete Logarithms. Let G = $g^r : 0 \leq r \leq n$ denote an n-element random cyclic group created by $g > 1$. Find the integer r = log(s) from a primitive element g and a random element $s = g^r G$.

2. The DH Problem in Computation Find g^{rs} for n, g, and a random pair of elements g^r

and g^s if you have $nandg$.

3. DH's choice Problem. Determine whether $g^t = g^{rs}$ if you have n, g, and a random triple of components g^r, g^s, and g^t.

4. Inappreciable Assumption. In polynomial time, a randomly generated triple (g^r, g^s, g^{rs}) $\in GxGxG$ is indistinguishable from a randomly picked triple (x, y, z) $GxGxG$.

Intractable tasks include computing the discrete logarithm on a random cyclic group and similar difficulties. The quickest algorithms execute in $O(n^{1/2+\in})$ exponential times, $\in > 0$. The number of group operations necessary to determine a single discrete logarithm instance is measured by the running time complexity.There has been many research on this problem over years and none has changed the time complexity of the discrete logarithm in random cyclic groups [book()].

4.3 Attacks on Discrete logs

It is simple to prove that several of these assaults are comparable to computing discrete logarithms over GF (p). Breaking the signature scheme is not yet comparable to computing discrete logarithms or breaking the distribution scheme.None of the attacks stated in this section, however, appear to be successful in breaking the system.Two groups of attacks will take place. We illustrate various methods for retrieving the secret key x in the first group, and some techniques for forging signatures without recovering x in the second group.

4.3.1 Attacks Aiming to Recover x

Attack 1: Given $m_i i = 1, 2, ..., l$ information compiled with the corresponding signatures $(r_i, s_i) : i = 1, 2, ..., l$ if an intruder tries to compute l equation of the form $m \equiv xr + ksmod(p-1)$.There are $l+1$ unknowns,The number of solutions to the system of equations is vast, and not all have been solved. Because each value for x results in a system of linear equations with a diagonal matrix of coefficients, each value for x offers a solution for the k_i.when $p-1$ is taken to have at least one large prime factors q,then recollecting $xmodq$ needs an exponential number of message-signature pairs. Note: suppose any k is used twice in signing,then the system of equation is singly computed and x can be retrieved.For the system to be secure no value of k should be used twice.

Attack 2: Trying to compute equations of the form $a^m \equiv y^r r^s modp$ always the sames as solving discrete logarithms over $GF(p)$, when both unknowns appear in the exponent.

Attack 3: Then the intruder will try to figure out and develop some linear dependencies in the unknown variables $k_i, i = 1, 2, ..., l$.This is the same as solving discrete logarithms

since if $k_i \equiv ck_j mod(p-1)$ hence $r_i \equiv r_j^c modp$.if c can be solved then solving discrete logarithms is easy.

4.3.2 Attacks for Forging Signatures

A forger might try to obtain r, s in order to satisfy given a document m. Computing s is analogous to solving a discrete logarithm problem over GF if $r = a^j modp$ is constant for some j selected at random (p). If the forger first fixes s, then the equation can be used to calculate r.

$r^s y^r = Amodp.$

Attack 4: Although it has not yet been proven that solving equation $r^s y^r = Amodp$ for r is at least as difficult as computing discrete logarithms, we believe that solving $r^s y^r = Amodp$ in polynomial time is not possible. The reader is challenged to come up with a polynomial time algorithm for addressing the problem $r^s y^r = Amodp$.

Attack 5: (3) appears to be solvable for both r and s at the same time, but we haven't been able to identify an efficient solution to do so.

Attack 6: The signature system provides for the following attack: an intruder can manufacture other genuine signatures and messages by knowing one legitimate signature for one message.This is because this technique prevents an intruder from signing any communication, the system is not broken. This property can be prevented by either ensuring that m have a specific structure or applying a one-way function on the message m before signing it. Because this technique prevents an intruder from signing any communication, the system is not broken. This property can be prevented by either ensuring that m have a specific structure or applying a one-way function on the message m before signing it.

Given a signature for a message (m),then

$$a^m = y^r r^s modp$$

Taking an integer A,B and C arbitrarily such that (Ar -Cs) is relatively prime to p - 1.then set

$$r^{'} \equiv r^A a^B y^C modp,$$

$$s^{'} \equiv sr^{'}/(Ar - Cs)mod(p - l)$$

,

$$m^{'} \equiv r^{'}(Am + Bs)/(Ar - Cs)mod(p - 1)$$

. since it is valid that $(r^{'}, s')$ signs the message $(m^{'})$.

solve

$$y^{r^{'}} r'^{s^{'}} \equiv y^{r^{'}} \left(r^{A} \alpha^{B} y^{C}\right)^{sr'/(Ar-Cs)}$$

$$y^{r^{'}} r'^{s^{'}} \equiv \left(y^{r'Ar-r'Cs+r'Cs} r^{Asr'} \alpha^{Bsr'}\right)^{1/(Ar-Cs)}$$

$$y^{r^{'}} r'^{s^{'}} \equiv \left((y^{r} r^{s})^{Ar'} \alpha^{Bsr'}\right)^{1/(Ar-Cs)}$$

$$y^{r^{'}} r'^{s^{'}} \equiv \alpha^{m'}$$

$$r' \equiv \alpha^{B} y^{C} \bmod p,$$

$$s' \equiv -r'/C \bmod (p-1),$$

$$m' \equiv -r'B/C \bmod (p-1).$$

It can be shown that (r', s') signs (m').

4.3.3 Baby Step, Giant Step

Shanks established the first attack on the discrete logarithm issue, which requires approximately $\sqrt[1]{n}$ steps and around $\sqrt[1]{n}$ storage to perform well for moderately big n. The following is Shanks' algorithm, which is now known as the Baby Step, Giant Step algorithm:

1. Put an integer $m \geq \sqrt[1]{n}$ and evaluate mP.
2. create and save a list of i_P for $0 \leq i \leq m$
3. Find the point $Q = jmP$ for $j = 0, 1..., m-1$ till one corresponds with a point from the save list.
4. If $i_P = Q - jmp$, then $Q = kP$ with $k = (i + jm)(modn)$.Hence $Q = i_P + jmP = (i + jm)P = kp$.

If we put an integer $m\sqrt[1]{n}$,then $m^2\sqrt[1]{n}$ this algorithm will work suppose that the solution k is such that $0 \leq k < m^2$.Taking $k = k_0 + mk$ where $k_0 \equiv k(modm)$ and $k_1 = (k - k_0)/m$.when

$i = k_0 and j = k_1$ respectively then $0 \leq k_1 < m$

We get

$$Q - jmp = Q - k_1 mp = kP - k_1 mP = (k - k_1 m)P = k_0 P$$

then the point corresponds with another point.

Assuming we did not know $n = |G|$, only an upper bound,then, for elliptic curve over F_q, we could apply this method with $m^2 \geq q + 1 + 2\sqrt{q}$ by Hasse's theorem.

Example. Let $G = E(F_{41})$, where E is given by $y^2 = x^3 + 2x + 1$. Let $P = (0, 1)$ and $Q = (30, 40)$. By Hasse's theorem,

suppose

$$|41 + 1 - E(F_{41})| \leq 2\sqrt{41} \implies 29.194 \leq E(F_{41}) \leq 54.806$$

, if $n = |G| \leq 54$. then we let $m = 8$ since $m \geq \sqrt{54}$. Points P_i for $1 \leq i \leq 7$ are

$$(0, 1), (1, 39), (8, 23), (38, 38), (23, 23), (20, 28), (26, 9).$$

When $Q = jmp for j = 0, 1, 2$ we evaluate to obtain

$$(30, 40), (9, 25), (26, 9),$$

Where we get a corresponding point to another point, since $j = 2$ corresponds with 7P, we then get

$$Q = (30, 40) = 7P + 16P = 23P$$

.

4.3.4 Pollard's ρ Method

The Baby Step, Giant Step algorithm has a significant drawback in that it necessitates a large amount of storage. Pollard's method takes about the same amount of time as Baby Step, Giant Step, but requires much less storage.Assume G is a finite group with $|G| = n$. Select a function $f : G \implies G$ that acts erratically. We begin with P0 and compute iterations, $P_{i+1} = f(P_i)$. There will be some indices i0 j0 such that $P_{i0} = P_{j0}$ since $|G| = n$. Then,

$$P_{i0+1} = f(P_{i0}) = f(P_{j0}) = P_{j0+1}$$

For every $\rho \geq 0$ $P_{i0+\rho} = P_{j0+\rho}$. As a result, $P_{i i \in N}$ has a periodic period of $j_0 - i_0$ or is a divisor of $j_0 - i_0$. Pollard's method gets its name from the fact that the diagram representing the procedure looks like a Greek letter. We should expect to find a match with j0 at most a constant times p if f is a randomly chosen random function,Then, at most a constant times $\sqrt{n}$, we should obtain a match with j_0.

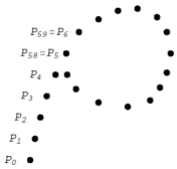

Figure 4.1: Pollard's ρ Method

Until a match is found, one implementation of the technique keeps all the points P_i. This requires around $\sqrt{n}$ storage, which is comparable to Baby Step, Giant Step. However, with a little extra computation, it is possible to perform a lot better.The concept is that once two indices differing by d match, all subsequent indices differing by d will be matches as well. Thus, for $i = 1, 2, 3, ...$ we can compute pairs (P_i, P_{2i}), but we only keep the current pair. The prior pairs will not be kept.

$$P_{i+1} = f(P_i) and P_{2(i+1)} = f(f(P_{2i}))$$

If $i > i_0$ and i is a multiple of d, then 2_i and i are a multiple of d apart. As a result, $P_i = P_{2i}$ is a match. There is a match for $i < j_0$ since $d \leq j_0$. As a result, the number of steps required to discover a match should be at most a constant multiple of $\sqrt{n}$.

The issue is determining which function f is appropriate. Divide G into s distinct subsets $S_1, S_2, ...S_s$ of roughly equal size. 2_s randomly generated integers $a_i, b_i (mod n)$.

$$M_i = a_i P + b_i Q$$

deifne

$$f(g) = g + M_i$$

if $g \in S_i$

Choose two random integers, a_0 and b_0, and set the starting point at $P_0 = a_0 P + b_0 Q$. We keep track of how these points are expressed in terms of P and Q when computing P_j. If

$$P_j = u_j P + v_j Q and P_{i+1} = P_j + M_i$$

, then

$$P_{j+1} = u_jP + v_jQ + a_iP + b_iQ = (u_j + a_i)P + (v_j + b_i)Q$$

so

$$(u_{j+1}, v_{j+1}) = (u_j + v_j) + (a_i, b_i).$$

finding a match $P_{i0} = P_{j0}$, then

$$u_{j0}P + v_{j0}Q = u_{i0}P + v_{i0}Q \implies (u_{i0} - u_{j0})P = (v_{j0} - v_{i0})Q$$

so $gcd(v_{j0} - v_{i0}, n) = d$, then

$$k \equiv (v_{j0} - v_{i0})^{-1}(u_{i0} - u_{j0})(modn/d).$$

This results in d options for k. Because d is usually small, we can test all alternatives until $Q = kP$. Example Let G = E(F1093) where E is the elliptic curve $y^2 = x^3 + x + 1$. using $s = 3$. Let $P = (0, 1)$ and $Q = (413, 959)$. It can be shown that $|P| = 1067$ and we want to find $k \in Z$ such that $Q = kP$.

Let $P_0 = 3P + 5Q$, $M_0 = 4P + 3Q$, $M_1 = 9P + 17Q$, and $M_2 = 19P + 6Q$. Let $f : E(F_{1093}) \implies E(F_{1093})$ be defined by

$$f(x, y) = (x, y) + M_i$$

if $x \equiv i(mod3)$.

In this case, x is an integer with the value $0 \leq x \leq 1093$. It is also possible to define $f(\infty) = \infty$. If we find $P_0, P_1 = f(P_0, P_2 = f(P_1)$ we then get

$$P_0 = (326, 69), P_1 = (727, 589), P_2(560, 365), P_3(1070, 260)$$

,

$$P_4 = (473, 903), P_5 = (1006, 951), P_6 = (523, 938), ...$$

$$P_{57} = (895, 337), P_{58} = (1006, 951), P_{59} = (523, 938), ...$$

S As a result, $P_5 = P_{58}$, and we keep track of the P and Q coefficients we discover. $P_5 = 88P + 46Q$ and $P_{58} = 685P + 620Q$. Then

$$P_{58} - P_5 = 567P + 574Q = \infty$$

$$Hence|P| = 1067$$

We solve

$$-574^{-1}597 \equiv (mod1067)$$

Hence, Q=499P so that k=499

.

4.4 Elliptic Curve Domain Parameters

There are two sorts of elliptic curve domain parameters: those over F_p and those over F_{2m}. Elliptic curve domain parameters over F_p are described in Section 3.1.1, and elliptic curve domain parameters over F_{2m} are described in Section 3.1.2.Elliptic curve domain parameters can be verifiably random, meaning that they are derived in part from the output function applied to a seed value S. Although verifiably random elliptic domain parameters are preferred, other options may be utilized for a variety of reasons, including improved performance.

4.4.1 Elliptic Curve Domain Parameters over Fp

Elliptic curve domain parameters over a finite field say F_p are a sextuple:

$$T = (p, a, b, G, n, h)$$

Taking an integer p and setting out the finite field F_p, two point $a, b \in F_p$ so setting out an elliptic curve $E(F_p)$ is defined by:

$$E : y^2 \equiv x^3 + ax + b(modp)$$

Considering a base point $G = (xG, yG)$ on $E(F_p)$, a certain prime called n which is the order of G, and an integer h that is the cofactor $h = E(F_p)/n$.

4.4.2 Recommended Parameters secp192k1

The sextuple $T = (p, a, b, G, n, h)$ specifies the elliptic curve domain parameters over F_p associated with a Koblitz curve secp192k1, where the finite field F_p is described by:

$$p = FFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFEFFFFFEE37$$

$$= 2^{192} - 2^{32} - 2^{12} - 2^8 - 2^7 - 2^6 - 2^3 - 1$$

The curve E: $y^2 = x^3 + ax + b$ over F_p is defined by:

$$a = 0000000000000000000000000000000000000000000000000$$

$$b = 0000000000000000000000000000000000000000000000003$$

We have the base point G in a compressed form is:

$$G = 03DB4FF10EC057E9AE26B07D0280B7F4341DA5D1B1EAE06C7D$$

this is the uncompressed form is:

$$G = 04DB4FF10EC057E9AE26B07D0280B7F4341DA5D1B1EAE06C7D9B2F2F6D9C5628A7844163D01$$

the order n of G and it cofactor are:

$$n = FFFFFFFFFFFFFFFFFFFFFFFEE26F2FC170F69466A74DEFD8D$$

$$h = 01$$

For all elliptic curve cryptography, you need $p =$ the prime, a and b which define the curve, $G =$ the generator point, $n =$ number of point in the field and $h =$ cofactor.

using weistress equation :

$$y^2 = x^3 + ax + b$$

To find the private key to an elliptic curve.The figure below shows us how the private key with the knowledge of elliptic cure and algorithm.

Using Python:

The result to the figure above is shown below:

Spyder (Python 3.7)

File Edit Search Source Run Debug Consoles Projects Tools View Help

C:\Users\hp\Documents

Editor - C:\Users\hp\Documents\king.py

king.py

```python
# secp256k1 domain parameters
Pcurve = 2**256 - 2**32 - 2**9 - 2**8 - 2**7 - 2**6 - 2**4 -1
Acurve = 0;
Bcurve = 7;
Gx = 0x79BE667EF9DCBBAC55A06295CE870B07029BFCDB2DCE28D959F2815B16F81798
Gy = 0x483ADA7726A3C4655DA4FBFC0E1108A8FD17B448A68554199C47D08FFB10D4BB
GPoint = (int(Gx),int(Gy)) # This is our generator point
N=0xFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFEBAAEDCE6AF48A03BBFD25E8CD0364141 # Number of points in the field
#Replace with any private key
privKey = 0xA0DC65FFCA799873CBEA0AC274015B9526505DAAAED385155425F7337704883E #replace with any private key

def modinv(a,n=Pcurve): #Extended Euclidean Algorithm/'division' in elliptic curves
    lm, hm = 1,0
    low, high = a%n,n
    while low > 1:
        ratio = high/low
        nm, new = hm-lm*ratio, high-low*ratio
        lm, low, hm, high = nm, new, lm, low
    return lm % n
def ECadd(a,b): # Not true addition, invented for EC. Could have been called anything.
    LamAdd = ((b[1]-a[1]) * modinv(b[0]-a[0],Pcurve)) % Pcurve
    x = (LamAdd*LamAdd-a[0]-b[0]) % Pcurve
    y = (LamAdd*(a[0]-x)-a[1]) % Pcurve
    return (x,y)
def ECdouble(a): # This is called point doubling, also invented for EC.
    Lam = ((3*a[0]*a[0]+Acurve) * modinv((2*a[1]),Pcurve)) % Pcurve
    x = (Lam*Lam-2*a[0]) % Pcurve
    y = (Lam*(a[0]-x)-a[1]) % Pcurve
    return (x,y)
def EccMultiply(GenPoint,ScalarHex): #Double & add. Not true multiplication
    if ScalarHex == 0 or ScalarHex >= N: raise Exception("Invalid Scalar/Private Key")
    ScalarBin = str(bin(ScalarHex))[2:]
    Q=GenPoint
    for i in range (1, len(ScalarBin)): # This is invented EC multiplication.
        Q=ECdouble(Q);
        if ScalarBin[i] == "1":
            Q=ECadd(Q,GenPoint);
    return (Q)
print; print ("************ Private key Generation **********");
print
PublicKey = EccMultiply(GPoint,privKey)
print ("the private key :");
print (hex(privKey));
print;
```

Figure 4.2:

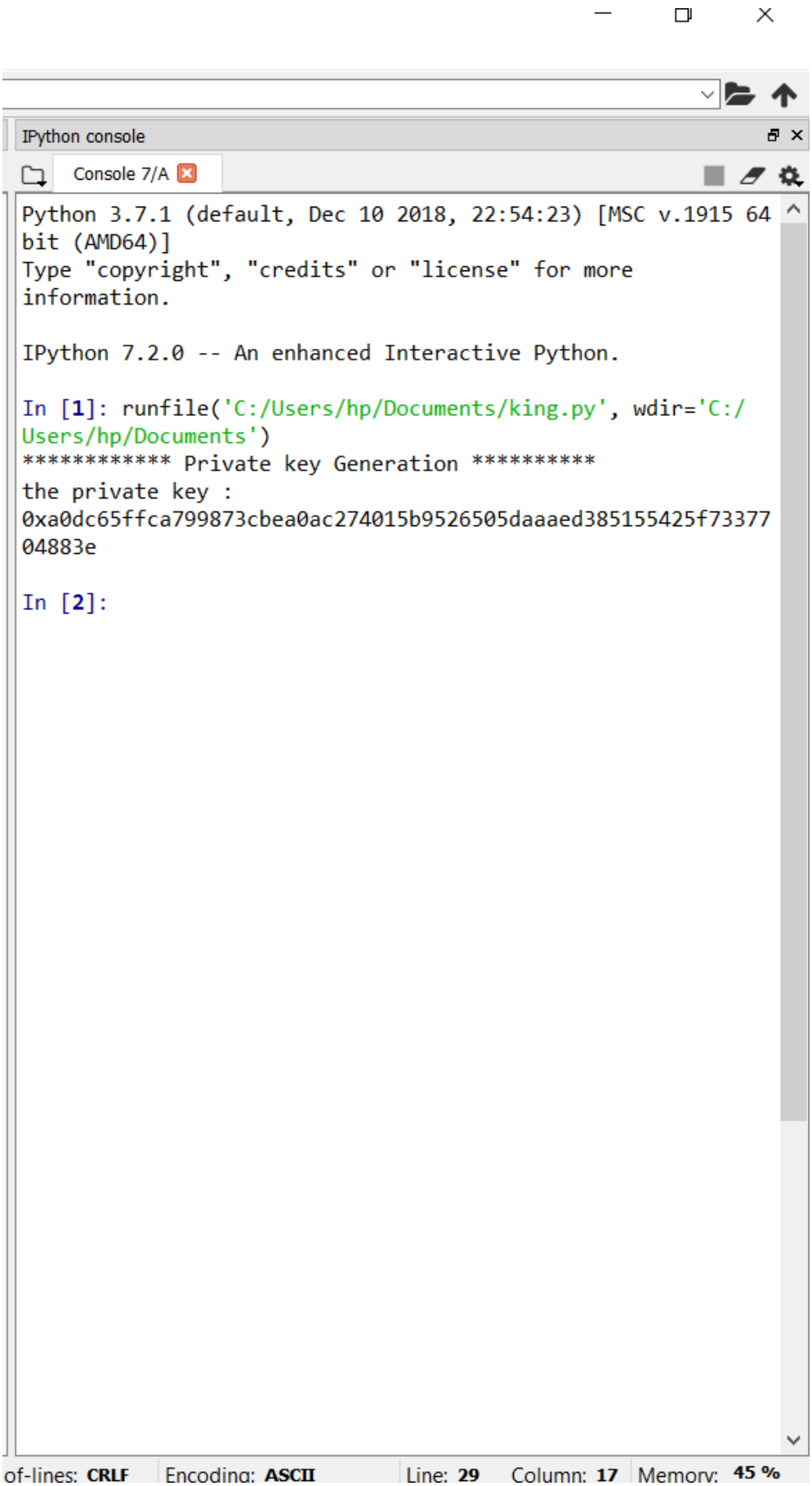
IPython console
Console 7/A
Python 3.7.1 (default, Dec 10 2018, 22:54:23) [MSC v.1915 64 bit (AMD64)]
Type "copyright", "credits" or "license" for more information.
IPython 7.2.0 -- An enhanced Interactive Python.
In [1]: runfile('C:/Users/hp/Documents/king.py', wdir='C:/Users/hp/Documents')
********** Private key Generation **********
the private key :
0xa0dc65ffca799873cbea0ac274015b9526505daaaed385155425f7337704883e
In [2]:
of-lines: CRLF
Encoding: ASCII
Line: 29
Column: 17
Memory: 45 %

Figure 4.3:

Chapter 5

Conclusion and Recommendation

5.1 Conclusion

Victor Miller and Neal Koblitz independently conceived the notion of utilizing elliptic curves as the basis of a group for the discrete logarithm problem in 1985, and this was the beginning of elliptic curve cryptography.The fact that no sub exponential technique existed to solve the discrete logarithm issue on a suitably designed elliptic curve is the major reason behind Elliptic curve cryptography's appeal. This meant that Elliptic curve cryptography used substantially lower parameters while maintaining the same level of security.Elliptic Curves provide security in the same way as traditional systems (such as RSA) do, but with fewer bits.

The use of elliptic curves in cryptography necessitated a smaller chip size, lower power consumption, and increased speed. In the contest of generating key, the cyclic group,Key pair generation together with the methods of application of rational numbers on elliptic curves in cyptography and finding point algebraically were used.

We also discussed about Diffie - Hellman Key Exchange on elliptic curve cryptography, ELGammal Public Key Encryption and ElGammal Digital Signatures which were geared towards the secure transfer of information. It was also found that, in channeling attacks and security twist the methods used were based on Attacks on Discrete logs, Baby Step, Giant Step, Pollard's p Method. A script was derived for generating private key cryptography.

5.2 Recommendation

Elliptic curve cryptography is considered to be a tool for Mathematical studies, specifically in getting to know, private key cryptography in ECC to keep confidential information between two parties along Channel attacks and twist-security attacks a based on the details and illustrative codes in this research.

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